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Virus Inactivation by Solvent/Detergent Treatment and the Manufacture of SD-Plasma

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Abbreviations: BVDV: bovine viral diarrhea virus,
CID_{50}: Chimpanzee infectious dose,
$TICD_{50}$: tissue culture infectious dose

Abstract

Solvent/Detergent (SD) is an extraordinarily effective means for eliminating enveloped viruses from plasma and plasma products [1]. The safety margin suggested by the rapid and complete kill of enveloped viruses observed in the laboratory has been confirmed repeatedly by groups worldwide and by thirteen years of routine clinical use encompassing an estimated 35 million doses of a wide variety of products. Throughout this time, there has not been a single documented case of enveloped virus transmission by an SD-treated product. This record of safety spawned the development of SD-treated plasma as a substitute for fresh frozen plasma (FFP) [2] and has encouraged the adoption of SD for the treatment of non-blood products such as monoclonal antibodies and those derived from recombinant DNA procedures [3]. This review summarizes the use of SD treatment, including its use in combination with other viral elimination procedures, and also summarizes Vitex's initial experience in the manufacture of SD-Plasma, recently licensed by the U.S. FDA.

Reported Usage of SD-Treated Products

The SD viral inactivation technology was first licensed for the treatment of clotting factor concentrates in 1985, and since that time has become the most widely used virucidal method for the treatment of plasma protein products around the world today. There is an abundance of laboratory, clinical and market use data demonstrating the efficacy of the SD process as a means of eliminating lipid-enveloped virus transmission by plasma derivatives. Viral safety has been demonstrated in numerous in vitro and in vivo studies, including seven different studies evaluating hepatitis virus inactivation in chimpanzees, and clinically confirmed in at least eleven independent clinical trials involving 427 hemophiliacs believed to be susceptible to hepatitis viruses and 455 hemophiliacs believed to be susceptible to HIV [1]. In addition, plasma from HIV-infected homosexual men has been subjected to SD treatment prior to fractionation to produce a hyperimmune anti-HIV immunoglobulin preparation. This treated plasma and the SD-treated immunoglobulin did not transmit HIV or hepatitis to chimpanzees and the immunoglobulin has proven to be safe in clinical trials [4]. Importantly, product safety was demonstrated prior to the time when screening tests for HIV or HCV became available [5].

Based on New York Blood Center records, approximately 35 million **doses** of SD-treated products have been administered worldwide (Table 1) [6] with no reported transmission of enveloped viruses and no reported toxicities. In terms of hemophilia care, assuming an average hemophiliac is treated with 100,000 IU of factor VIII or factor IX per year, the quantity of product transfused exceeds 160,000 man-years of treat-

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ment. This safety record indicates that SD treated products are far safer than any non-virally inactivated blood component. As compared with a similar analysis prepared one year earlier, there is increased usage of SD in the preparation of immune globulins including IVIG, virally inactivated transfusion plasma, and topical hemostatic agents (e.g. thrombin and fibrin glue). Use in the preparation of immune globulins follows the documented transmission of hepatitis C by a non-virally inactivated IVIG preparation [7].

Table 1. SD-Treated Product Usage: 1985-1997

Product	Units	Doses (approx)
FVIII	15,814 MU	15,814,000
FIX	877 MU	877,000
Prothrombin complex	194 MU	194,000
vWF	2,832 MU	2,832,000
FVII + FVIIa	77.2 MU	77,000
Fibrinogen	219,656 g	54,900
Other (Prot C, FXI, ATIII)	189 MU	63,000
Plasma	3,010,000 units	602,000
Thrombin	28,017 MU	5,603,000
Fibrin Glue	399,150 ml	200,000
IMIG & IVIG	28,951,000 g	5,790,000
Anti-D IgG	328,936 mg	1,096,000
HB-Ig	466,115 ml	117,000
CMV-Ig	845,895 g	121,000
RSV-Ig	496,807 g	199,000
Other Ig (anti-tetanus, etc)	971,000 ml	486,000
MAB (anti-platelet, anti-cancer, etc)	1,121,940 vials	1,122,000
	Sum:	**35,248,000**

Combination of SD with Other Viral Elimination of Procedures

Documented viral transmission by virally inactivated blood plasma products has been rare. Nonetheless, the transmission of HAV by certain SD-treated coagulation factor concentrates [8] and the transmission of hepatitis by certain heat-treated coagulation factor concentrates [9] have encouraged the adoption of so-called "double viral elimination" procedures. Enhanced safety can be achieved through protein purification methods, such as antibody-mediated affinity chromatography [10, 11], incorporation of "nanofilters" with sufficiently small pore size so as to trap viruses while permitting the passage of the desired proteins [12], or the incorporation of a second viral inactivation step [13, 14, 15]. Each approach has certain advantages and certain limitations (Table 2).

Table 2. Double Viral Elimination

Approach	Example	Comment
Monoclonal antibody affinity purification	factor VIII	Eliminate ≥4 logs of variety of viruses. Use restricted to products where single pure protein is required and to products where a dose is in the milligram or submilligram range.
Nanofiltration	factor IX	Eliminate ≥4 logs of variety of viruses. Use restricted to proteins of lower molecular weight (e.g. ≤150,000 daltons). Elimination of smaller viruses (e.g. HAV, parvovirus) requires an effective pore size of ≤15nm.
SD + Heat	factor VIII, fibrinogen	Further assures enveloped virus safety and increases spectrum of susceptible viruses. Non-enveloped viruses also tend to be heat stable. One implementation was found to be immunogenic [15].
SD + UVC	fibrin glue	Both enveloped and non-enveloped viruses killed to high degree. Requires specialized equipment. Still in clinical trials.

SD-Plasma

The viral safety of transfused blood products currently depends on the dual safeguards of donor selection and viral screening of donations. Despite these efforts, there remains a threat to the blood supply by the donation of blood by seronegative donors during the infectious window period when the donors are undergoing seroconversion. A recent study [16] suggests that among donors whose blood passed all screening tests, the risks of giving blood during an infectious window period were estimated as follows: for HIV, 1 in 493,000; for HTLV, 1 in 641,000; for HCV, 1 in 103,000; and for HBV, 1 in 63,000. The authors calculated the aggregate risk to any of these viruses at 1 in 34,000 per unit transfused. Since most patients who receive fresh frozen plasma (FFP) are exposed to 2-8 donors and some patients, notably liver transplantation patients, plasma exchange therapy patients, and many congenital coagulation factor-deficient patients are exposed to many times this number, the aggregate risk to the patient could approximate 0.1-1%. New screening tests that shorten the window periods for these four viruses should reduce the risks to some extent, but will in no way eliminate the residual risks of transfusion. For example, viral mutants (e.g. HIV type O and a new form of HBV [17]) which are not detected by current tests are being discovered, and the non-window period risk for HCV transmission is estimated at 20% of the total [18].

Residual viral risks have been addressed successfully in other plasma products through implementation of viral inactivation procedures. Until recently, FFP was the sole plasma product not available in a virally inactivated form in the U.S. SD-Plasma is designed to eliminate these risks. The safety of SD-Plasma derives from two factors: SD treatment protects against all enveloped viruses and the guaranteed presence of antibody associated with pooling protects against hepatitis A virus (HAV) and parvovirus B19 (B19), the two non-enveloped viruses reported in a transfusion setting.

Viral inactivation:

The method of SD treatment adapted for whole plasma involves incubation of pooled thawed plasma from volunteer donors with the organic solvent tri(n-butyl) phosphate (TNBP) and the detergent Triton X-100 for 4 hours at 30°C. As prepared at Vitex, individual pooled lots contain plasma from no more than 2,500 donors, or 1/10 to 1/40 the number employed in pools for plasma fractionation. SD treatment results in rapid viral killing, with complete kill typically observed within the first 15 minutes of the total 4 hour treatment time (Figure 1).

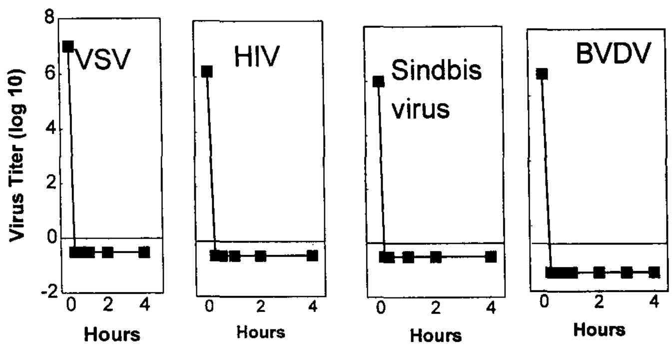

Fig. 1 Rate of inactivation of viruses on treatment of plasma with SD. Each of the listed viruses were added to plasma followed by 1% TNBP and 1% Triton X-100. The plasma was mixed continuously at 30°C. Samples were taken at the indicates: time points and assayed for residual, infectious virus *in vitro.*

Studies have been performed with human pathogenic viruses and laboratory viruses which are models of human disease, Sindbis virus and BVDV which are models for HCV, and duck HBV, a model for human HBV. We have demonstrated inactivation of $\geq 10^{7.5}$ $TCID_{50}$ VSV, $\geq 10^{6.9}$ $TCID_{50}$ Sindbis virus, $\geq 10^{6}$ CID_{50} of HBV, $\geq 10^{6.1}$ CID_{50} of BVDV, $\geq 10^{5}$ CID_{50} of HCV, $\geq 10^{7.2}$ $TCID_{50}$ of HIV, and $\geq 10^{7.3}$ ID^{50} of duck HBV.

To further assess the robust nature of SD treatment, we studied the kinetics of inactivation of BVDV (selected because it is a model of HCV) with purposeful reductions in the concentration of TNBP and Triton X-100 (Table 3). Viral kill was complete (≥ 5 $\log_{10}$) by the first time point taken (15 minutes) even when the concentrations of TNBP and Triton X-100 were reduced to 0.4% and 0.85%, respectively. Standard treatment conditions use 1% TNBP and 1% Triton X-100 for a minimum of 4 hours.

Table 3. Effect of Reduced Levels of TNBP and Triton X-100 on Inactivation of BVDV

Sample	Viral Titer ($TCID_{50}$)		
	1% TNBP + 1% Triton X-100	0.8% TNBP + 0.85% Triton X-100	0.4% TNBP + 0.85% Triton X-100
Minus SD; 240 min	6.07	5.53	5.83
Stop control	5.95	5.89	5.95
Plus SD; 15 min	≤0.87	≤0.87	≤0.87
Minus SD; 0 min	5.95	6.31	5.95
Plus SD; 30 min	≤0.87	≤0.87	≤0.87
Plus SD; 60 min	≤0.87	≤0.87	≤0.87
Plus SD; 120 min	≤0.87	≤0.87	≤0.87
Plus SD; 240 min	≤0.87	≤0.87	≤0.87

Antibody presence

Because of pooling, the presence of protective antibodies to HAV and B19 is guaranteed. With both HAV and B19, antibodies have proven to be effective in the prevention of transmission as well as the abrogation of established disease. As applied to SD-Plasma, if virus is present in the initial pool, it would be complexed and neutralized by antibody. For virus which escaped full neutralization, or which failed to be removed by the reticuloendothelial system, the co-administration of free antibody would be expected to prevent clinical manifestation of the disease. Although transmission of HAV by coagulation factor products has been reported, transmissions are rare and appear to be related to the more vigorous methods of purification employed today which deplete the product of antibody. This problem does not exist for SD-Plasma. The first 34 lots of SD-Plasma produced at Vitex were shown to contain a minimum of 0.8 IU/mL of anti-HAV antibody and 8 U/mL of anti-B19 antibody. Thus, recipients of SD-Plasma (60 kg; 10 ml/kg) will receive a minimum of 5,000 mg of IgG, 480 IU of anti-HAV, and 30-times the quantity of antibody currently prescribed as HAV prophylaxis for travelers. Similarly with B19, a dose of SD-Plasma contains almost as much anti-B19 antibody as a dose of IVIG used therapeutically to treat a chronic B19 infection [19]. In the laboratory, a 20- to 50-fold dilution of plasma protects cells in tissue culture against infection [20]. It is interesting to note that each of the first 34 lots of SD-Plasma were negative when tested for HAV gene sequences by PCR, using a test capable of detecting as few as 10 copies/mL.

Coagulation factor content

The coagulation factor content of the first 34 lots of SD-

Table 4. SD-Plasma Final Product (n = 34)

	Factor V (u/ml)	Factor VII (u/ml)	Factor X (u/ml)	Factor XI (u/ml)	Factor XIII (u/ml)	FBG (mg/ml)
Average	0.83	1.01	1.06	1.04	1.08	2.67
SD	0.06	0.10	0.04	0.09	0.09	0.17

Placed by Vitex is given in Table 4; each of the tested coagulation factors must exceed 0.7 u/ml (1.7 mg/ml for fibrinogen) for the product to be released. All of the tested coagulation factors exceed this minimum, and the small standard deviation indicates consistency in manufacture, and are generally reflective of the European experience in manufacturing SD-Plasma [21, 22].

TNBP and Triton X-100 content

All but 3/34 lots had undetectable levels of TNBP (≤2 ppm) and of Triton X-100 (≤1 ppm). One lot had 2.1 ppm of TNBP and two lots had 1-2 ppm of Triton X-100. These low levels further assure the safety of SD-Plasma.

Conclusion

Licensure of SD-Plasma by the FDA and other regulatory bodies has transformed FFP into a modern, sterile pharmaceutical, available in standardized doses and containing a uniform content of coagulation factors and other valuable proteins. Importantly, patients who are transfused with plasma no longer have to rely on screening tests alone for viral safety. These patients can now enjoy the same safety increment provided by viral inactivation as that provided to all other plasma protein products. All available evidence indicates that SD treatment eliminates the risk of transmission of the major pathogenic viruses in blood, HIV, HBV, and HCV, as well as other enveloped viruses of less clear importance currently, including HGV. The guaranteed presence of antibodies to HAV and B19 provide further protection. Confidence in the viral safety of SD-Plasma derives from the extensive experience by numerous laboratories validating the SD method and from 13 years of routine human use. The viral safety of SD-Plasma is of special importance in the treatment of congenital factor deficient patients, who may require many plasma infusions over a lifetime; patients with TTP, who frequently receive hundreds of units of plasma; patients who are exchange transfused; and patients who wish to have the confidence that comes with an SD-treated blood product.

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